

Global Facilities - Design & Construction Standard - I&C - Instrumentation

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

This Standard defines the minimum requirements for design, supply, and commissioning of instrumentation in Micron greenfield project, including installation and cabling.

Any deviation from this standard shall be submitted for approval during evaluation. Deviation list format shall refer to Appendix A in this standard document.

2.0 Scope

2.1 Site(s) impacted

This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

2.2 Vendor Scope

2.2.1 Furnish, install, and leak test pneumatic tubing, piping and manifolds from the instrument air header isolation valve to the instruments and devices that require instrument air (.e.g. damper, control valves, etc). Provide a pressure gauge with pressure regulator or downstream so that you can determine the set pressure.

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- 2.2.2 Provide initial setup, calibration and testing of the instrumentation system including control valves, dampers, and in-line devices.
- 2.2.3 Provide instruments and control valves suitable for the application as required.
- 2.2.4 The design and selection shall refer to the Piping & Instrumentation Diagrams (P&IDs) and Sequence of Operation (SOO).
- 2.2.5 The package requirements shall include, but not limited to the following:
 - a. Instrument Datasheets, which indicate the instrument tag and associated project name.
 - b. Instrument List, with recommended format can refer to Appendix B in this standard document.
 - c. Bill of Quantity (BOQ)
 - d. Valve sizing, calculation, and selection list.
 - e. Calibration certificates.
 - f. Commissioning report

3.0 Definitions and Acronyms

Acronym	Definition / Description
A/D	Analog to Digital
AC	Alternating Current
BOQ	Bill of Quantity
D/A	Digital to Analog
DC	Direct Current
EHS	Environmental, Health, and Safety
EMI	Electro-Magnetic Interference
FAT	Factory Acceptance Test
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
FMCS	Facility Management and Control System
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team

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Acronym	Definition / Description
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
I&C	Instrumentation & Control
MAU	Makeup Air Unit
ORPORP – Oxidation	Oxidation-Reduction PotentialReduction Potential
P&ID	Piping and Instrument Diagram
PCW	Process Cooling Water
PLC	Programmable Logic Controller
RFI	Radio frequency interference
RTD	Resistance Temperature Device
SAT	Site Acceptance Test
SCADA	Supervisory Control and Data Acquisition
SOO	Sequence of Operation
T&H	Temperature and Humidity
ZRP	Zero Reference Pressure

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Applicable Codes and Standards

- a. American National Standard Institute (ANSI)
- b. International Organization for Standardization (ISO)
- c. Electric Industries Association (EIA)
- d. International Electrotechnical Commission (IEC)
- e. Instrument Society of America (ISA)
- f. National Electrical Code (NEC)
- g. National Electrical Manufacturers Association (NEMA)
- h. Underwriter Laboratories (UL)
- i. Japanese Industrial Standards (JIS)
- j. American Society for Testing and Materials (ASTM)
- k. China National Standard (CN)
- l. IEC 60051 Direct acting indicating, measuring instruments
- m. IEC 60417 Graphical symbols for use on equipment.
- n. IEC 60445 Identification of equipment terminals.
- o. IEC 60502 Cables.
- p. IEC 60529 Degrees of protection provided y enclosures (IP code).
- q. IEC 60654 Series: Operating Conditions for Industrial-Process measurement and control equipment.
- r. IEC 60801 EMI and RFI immunity.

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- s. IEC 61000 Electromagnetic compatibility (EMC).
- t. IEC 61285 Industrial-Process Control – Safety of analyzer houses.
- u. IEC 61131 programmable logic controllers.
- v. IEC 61508 All parts – functional safety of electrical/electronic/programmable electronic safety-related systems.
- w. IEC 61511 All parts – functional safety – safety instrumented systems for the process industry sector.
- x. ISA 5.1 Instrumentation symbols and identification
- y. ISA 5.4 Instrument loop diagrams.
- z. ISA 5.5 Graphic symbols for process displays.
- aa. ISA RP12.06.01 Recommended practice for wiring methods for hazardous.
- bb. ANSI/NFPA 75 Protection of electronic computer/data processing equipment.

6.0 Design Requirement

- 6.1 Provide latest revision of products that the manufacturer is currently manufacturing.

6.2 Instrument Tubing and Piping

Design pneumatic instrument systems and impulse lines for personnel safety and to facilitate calibration, maintenance, and replacement:

- 6.2.1 Design instrument systems with supports to ensure proper operation, including strain relief on connections to mitigate the consequences of material fatigue.
- 6.2.2 Provide tubing connections with adequate space for leak tests and replacement of fittings without disturbing adjacent tubing connections.
- 6.2.3 Provide instrument isolation valves, for instruments mounted on process lines, to facilitate isolation and removal of the instrument.

6.3 Devices:

- 6.3.1 Transmitters and measuring elements are compatible with the pressure and temperature ranges of process parameters and with processing piping specification.
- 6.3.2 Sensor-wetted parts compatible with the process fluids to be measured.
- 6.3.3 In-line component cleaning and packaging in compliance with the Process System requirements.

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- 6.3.4 Integral indicators where installed devices are visible to plant personnel.
 - 6.3.5 Enclosure ratings: Suitable for the service and location.
- 6.4 Fault Tolerant:
 - 6.4.1 Satisfactory operation without damage at 110% above and 85% below rated voltage and ± 3 Hz frequency variation.
 - 6.4.2 Static, transient short-circuit protection.
 - 6.4.3 Communication lines protected against incorrect wiring, static transients, and induced magnetic interference.
 - 6.4.4 Over/under range protection $\pm 50\%$.
- 6.5 Instrument analog signals for electronic instrument systems shall be 4-20mA_{dc}, loop powered when possible and utilize Hart protocol for configuration when available. If loop powered is not possible then isolation is required.
- 6.6 Generally, the instruments shall be electronic except for the final control valves and damper actuators which are typically pneumatic.
- 6.7 Generally, the transmitter will have local indicators. Local indicators shall be LCD type programmable displays with outputs scaled and displayed in the engineering units measured. Where the sensing element is required to be out of sight, an indicator or indicating transmitter shall be located in accessible level.
- 6.8 All instrument items shall be identified on P&ID.
- 6.9 Submittal requirements shall include, but not limited to:
 - a. Instrument List, with recommended format can refer to Appendix B in this standard document.
 - b. Instrument Datasheets, which indicate the instrument tag and associated project name.
 - c. Valve sizing, calculation, and selection list.
 - d. Instrument hook-up drawings.

7.0 Products

7.1 Analyzer Instrument Requirements.

7.1.1 Humidity

- 7.1.1.1 The humidity sub-category of analyzers includes those instruments that measure relative humidity for all or some portion of the full 0 to 100% Relative Humidity range; Dew Point analyzers that measure the Dew Point in either degrees Fahrenheit or degrees Celsius.
- 7.1.1.2 Relative Humidity Analyzers for rooms and ducts will generally measure a minimum range of 20% RH to 95% RH and be 1% or better accuracy in that range.
- 7.1.1.3 Relative Humidity Analyzers for installation immediately downstream of humidification media or steam bars shall generally measure a minimum range of 20% RH to 100% RH (condensing).
- 7.1.1.4 Dew Point Analyzer for rooms or ducts will generally be accurate to 1.0 °F or 0.5 °C.

7.1.2 PH Analyzer

- 7.1.2.1 pH Analyzer shall measure from 0 to 14 pH and be 0.01pH accuracy.
- 7.1.2.2 Sensor shall be temperature compensated to achieve the accuracy.
- 7.1.2.3 pH Analyzers may be of the insertion probe type or full immersion probe type such that they give a representative reading of the pH of the tank, vessel, or line that they are assigned to monitor. All mountings should be designed so that relocation of removal is possible.
- 7.1.2.4 Minimum conductivity of the measured steam shall be compatible with analyzer.
- 7.1.2.5 pH probes are to be installed in a slipstream of bypass to allow for maintenance of the device.
- 7.1.2.6 Sensor with automated cleaning is highly recommended at wastewater treatment plant.
- 7.1.2.7 Recommended manufacturers:
 - 1. Thornton (Mettler Toledo)
 - 2. M4 Knick
 - 3. Rosemount
 - 4. E&H
 - 5. Yokogawa
 - 6. Or Equivalent

7.1.3 Conductivity and Resistivity Analyzers

- 7.1.3.1 Conductivity/Resistivity sensors shall have a range appropriate to the application with the normal value at mid-point of range. Accuracy shall be 0.5% of full scale or better.
- 7.1.3.2 Sensor shall be temperature compensated to achieve the accuracy.
- 7.1.3.3 Conductivity and Resistivity analyzers shall be coordinated with the piping design of the area where they are to be located to insure proper installation.
- 7.1.3.4 Conductivity and Resistivity Analyzers may be of the insertion probe type or full immersion probe type depending on installation requirements. The probes shall be placed such that they give a representative reading of the conductivity/resistivity of the tank, or line that they are assigned to monitor.
- 7.1.3.5 Conductivity/Resistivity probes are to be installed in a slipstream or bypass to allow for maintenance of the device.
- 7.1.3.6 Recommended manufacturers:
 - 1. Thornton
 - 2. M4 Knick
 - 3. Hach
 - 4. Rosemount
 - 5. E&H
 - 6. Yokogawa
 - 7. Or equivalent

7.1.4 Oxidation-Reduction Potential (ORP) Analyzers

- 7.1.4.1 ORP Analyzers shall measure over a range of -500mV to +500mV and have an accuracy of 0.1%. Output Scale Expansion, zero suppression, up to ± 2550 mV.
- 7.1.4.2 ORP analyzers may be of the insertion probe type or full immersion probe type depending on installation requirements. The probes shall be placed such that they give representative reading of the ORP of the tank, vessel, or line that they are assigned to monitor.
- 7.1.4.3 ORP probes are recommended to be installed in a slipstream or bypass to allow for maintenance of the device.
- 7.1.4.4 Recommended manufacturers:
 - 1. Rosemount
 - 2. Thornton
 - 3. M4 Knick
 - 4. Or equivalent

7.1.5 Dissolved Oxygen (O2) Analyzers

- 7.1.5.1 Dissolved O2 analyzers shall be of the slip-stream type depending on installation requirement, and the probes shall be in the side stream to analyzer, with drain to DI reclaim. Bypass type installations are not permitted due to the sensor's lo-flow operation.
- 7.1.5.2 Dissolved O2 analyzers shall measure 0-100ppb, with 1% accuracy.
- 7.1.5.3 Recommended manufacturers:
 - 1. Thornton (Mettler Toledo)
 - 2. M4 Knick
 - 3. Orbisphere
 - 4. Swan Analytical Instruments
 - 5. Or equivalent

7.1.6 Total Organic Carbon (TOC) Analyzers

TOC analyzers for Ultra-Pure Water shall measure a range of 0.1 ppb to 500 ppb and have an accuracy of ± 1 ppb. For wastewater the range shall be 0 – 20 ppm.

Recommended manufacturers:

- 1. Anatel
- 2. GE Sievers
- 3. Or equivalent

7.1.7 Turbidity Analyzers

Turbidity analyzers for Ultra-Pure Water shall have a range of 0-20 NTU with an accuracy of ± 0.1 and for Waste Water Treatment shall have a range 0 – 1000 NTU with 0.1 NTU accuracy.

Recommended manufacturers:

- 1. Hach
- 2. E&H
- 3. Or equivalent

7.1.8 Chlorin Analyzers

Chlorin analyzers shall have a range of 0-20 ppm with an accuracy of ± 0.1 ppm.

7.2 Damper Actuators

- 7.2.1 Damper actuators shall be pneumatic spring-opposed diaphragm, spring-opposed rolling diaphragm, spring-opposed piston, rack and pinion (spring-opposed or double acting as required) piston, or Scotch-yoke type piston (for high-torque cases). Electric or Electro-hydraulic types may be considered only in special high-torque cases or in cases where instrument air is not readily available.
- 7.2.2 All actuators shall provide adequate torque to stroke and shut off the damper under the maximum differential pressure (+ 25% safety factor) to which the damper may be exposed.
- 7.2.3 Diaphragm actuators for proportional control shall be sized for 3 to 15 psig, unless the largest standard actuator is inadequate, in which case a pneumatic or electro-pneumatic positioner shall be used. The pneumatic pressure supply is 1.4 to 6 bars
- 7.2.4 Piston actuators may be specified as necessary. Piston operators shall be provided when diaphragm operator cannot develop enough force to operate the damper, insufficient room is available to accommodate an oversized diaphragm top works, or the diaphragm operator has insufficient stroke. If a piston-type is required, the actuators should be placed in a cleanroom air stream, if possible. Typical application of piston actuators shall be for exhaust and scrubber duct dampers.
- 7.2.5 If limit switches are required for the application, then specify integral limit switches if integral limit switches come available with the chosen actuator.
- 7.2.6 The following table shows the damper fail positions that shall be configured.

Table 1: Damper Fail Position

Control Damper	Control Damper Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
Process & General Exhaust Control Damper	Fail Last State	Fail Last State	Fail Last State

- 7.2.7 Any control damper actuators configured to fail-last state shall have the capability of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.
- 7.2.8 All control damper shall have a manual override bypass, to allow manual operation of the damper upon a failure.

7.3 Valve

7.3.1 Body material and rating shall conform to the piping requirement as a minimum:

- 7.3.1.1 Cast or forged carbon steel is generally used for non-corrosive applications with temperature above -20 °F or -30 °C and below 750 °F or 400 °C.
- 7.3.1.2 Alloy steel such as 316 stainless steel is generally used for clean water service (such as process cooling water), corrosive service, some flashing service, and for temperatures 750 °F (400 °C) and above, and -20 °F (-30 °C) and below.
- 7.3.1.3 Bronze is used only on water or air.
- 7.3.1.4 Plastic (PVDF, TFE) to be considered for ultra-pure water (UPW) service.
- 7.3.1.5 Plastic may be considered for certain corrosive services where temperatures and pressure permit.
- 7.3.1.6 Lined bodies may be considered for certain corrosive services where temperature and pressure permit.

7.3.2 Valve Connections shall conform to the following guide:

- 7.3.2.1 Flanged connections shall be used for all services with lines 2 inch or larger unless prohibited by the piping specifications. Butterfly control valves shall have lug-type flanges. Wafer style flanges are not acceptable. All non-flanged connections such as threaded shall be pre-approved through project Design team and Micron representative.
- 7.3.2.2 Threaded connections shall be used for lines less than 2 inch unless the piping specification prohibits threaded connections for particular service. Cold glycol and glycol hot water is a service that shall use flanged connections only, independent of valve size. The thread shall meet National Pipe Thread (NPT) dimensions.
- 7.3.2.3 Threaded valve installations shall have a union installed in the pipe to allow for removal of the valve.

7.3.3 Control valve body sizes shall normally be ½ inch minimum. Reduced ports shall be used as required. Body size shall not be more than two pipe sizes smaller than the line unless separately approved.

7.3.4 Valve type and size shall be selected taking into account such factors as cost, operating and design conditions, fluid type, rangeability required, allowable leakage, noise and any other special requirements.

7.3.5 Valve Actuators

- 7.3.5.1 Globe-style control valve actuators shall be the spring-opposed diaphragm or spring-opposed rolling diaphragm type.
- 7.3.5.2 Butterfly control valve actuators shall be spring-opposed diaphragm, spring-opposed rolling diaphragm, spring-opposed piston, rack and pinion (spring-opposed or double-acting as required) piston, Scotch-yoke type piston (for high-torque cases). Electric or electro-hydraulic types may be considered only in special high-torque cases.
- 7.3.5.3 All actuators shall be adequate to stroke the valve under the maximum differential pressure to which the valve may be exposed and still achieve full-closed and full-open positions.
- 7.3.5.4 Diaphragm actuators shall be sized for 3 to 15 psig signals, unless the largest standard actuator is inadequate, in which case a 6 to 30 psig spring and a valve positioner shall be used. The pneumatic pressure supply is 1.4 to 6 bars.
- 7.3.5.5 Piston operators shall be provided when the diaphragm operator cannot develop enough force to operate the valve, insufficient room is available to accommodate oversized diaphragms top works or the diaphragm operator has insufficient stroke. If piston-type are required, the valve should not be placed in a clean room air stream.
- 7.3.5.6 The following table shows the control valve fail positions that shall be configured.

Table 2: Control Valve Fail Positions

Control Valve	Control Valve Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
Dry Cooling Coil Temperature Control Valve	Fail Last State	Fail Last State	Fail Last State
Process Cooling Water Temperature Control Valve	Fail Last State	Fail Last State	Fail Last State
Make Air Unit Temperature Control Valve	Fail Last State	Fail Last State	Fail Last State

- 7.3.5.7 Any control valve actuators configured to fail-last state shall have the capability of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.
- 7.3.5.8 All control valves shall have a manual override bypass, to allow manual operation of the valve upon a failure.
- 7.3.6 Valve Positioners
 - 7.3.6.1 Recommended practice is positioners shall be provided on all valves 3 inch and above, and below 3 inch if the process requires.
 - 7.3.6.2 All positioners shall be a Smart type electro-pneumatic with Hart protocol and the capability to auto calibrate.
 - 7.3.6.3 Positioners shall have three gauges, unless they are electro-pneumatic positioners, in which case they shall have two gauges. The two gauges shall read air to the diaphragm and supply air pressures. The third gauge shall read instrument signal pressure if present.
- 7.3.7 Valve Sizing Considerations
 - 7.3.7.1 Valve sizing shall be based on selecting a valve that has the calculated Cv for maximum flow fall into a travel range from 50% to 80%. The calculated Cv for minimum flow (if given) shall fall into a travel position of greater than 10%. ISA S75.01 or the selected valve manufacturer's formula shall be used to determine valve Cv.
 - 7.3.7.2 This Cv shall be used to determine valve capacity. The Cv shall be calculated at minimum, maximum, and normal flow rate conditions, if these are given. For HVAC processes the flow rates often vary from zero up to the maximum, so only the maximum is given in many cases.
- 7.3.8 The noise generated by the control valve shall be calculated and entered on the control valve specification sheet. Where the calculated sound power level exceeds 85 dBA (29 CFR 1910.94) (measured at 3ft or 1m from the valve) steps shall be taken to reduce the noise. The preferred method is with valve selection rather than path treatment. Path treatment methods may be acoustic insulation, heavier walled pipe or in-line silencers.
- 7.3.9 Tight shutoff valves, except de-pressurizing valves, requiring tight shutoff seats shall normally use reinforced Teflon for the soft seats if Teflon is suitable to the process fluid. Valve leakage class should be specified on the data sheets. Valve leakage class designations shall conform to ANSI standard B16, 104 or FCI 70-2.

7.3.10 An anti-cavitation valve (high differential pressure control valve) will be utilized to provide smooth and accurate capacity control of high pressure drop liquids without cavitation, erosion and high noise levels, as approved by Design team and Micron representative.

7.4 Flow Measurement

7.4.1 General requirements

- 7.4.1.1 System shall be evaluated for the use of intrusive type flowmeters from a "system shutdown" perspective.
- 7.4.1.2 Flowmeters should be installed with the straight runs upstream and down per the manufacturer's recommendations.
- 7.4.1.3 The appropriate NEMA classification for the area in which it resides shall be met.
- 7.4.1.4 Flowmeters shall be able to meet the pressure and temperature ratings of the piping run in which they reside and be chemically compatible with the process fluid.
- 7.4.1.5 The flow transmitter is recommended to be two wire and provide a 4-20 mA dc linear output for a 0-100% flow span.
- 7.4.1.6 Switch provision shall be rated at a minimum of 5 amps at 120/240 Vac or 24 Vdc. The switch shall achieve a rating that meets the service for which they are intended.
- 7.4.1.7 Local display shall show current flow and totalization amount at minimum.

7.4.2 Magnetic Flowmeters

- 7.4.2.1 Magnetic flowmeters shall be flanged for sizes 2.5 inches and above. They may be flanged threaded or wafer for sizes 2 inches or less.
- 7.4.2.2 The conductivity of the fluid shall be compatible with the selected magnetic flowmeter. Many liquids, such as hydrocarbons or UPW water, are not conductive.

7.4.3 Ultrasonic flowmeters

- 7.4.3.1 For non-invasive flow measurement of liquids and gases in full closed systems, a clamp on ultrasonic flowmeter shall be utilized. Standard output signal of 4-20mA current linear to flow. Accuracy to $\pm 3\%$ of actual flow. For some applications, bi-directional flowmeters may be required.
- 7.4.3.2 For clean liquid media, transit-time technology should be used.

7.4.3.3 For dirty or aerated media, doppler technology should be used.

7.4.4 Orifice Plates and Flanges

7.4.4.1 Orifice plates and flanges shall be able to meet the pressure and temperature ratings of the piping run in which they reside.

7.4.4.2 The material of the orifice plates and flanges shall be compatible with the fluid to be measured and the piping classification of the run in which it is to reside.

7.4.4.3 Orifice plates shall be Stainless Steel SS316 unless that material is incompatible with the fluid to be measured.

7.4.4.4 Orifice plates shall have a tag with the upstream side stamped with the orifice bore, material, and tag number.

7.4.4.5 Orifice flanges shall be oriented such that their taps are at the side of the pipe.

7.4.4.6 Differential sensing lines shall have horizontal sections slope down towards the dP cell transmitter.

7.4.4.7 The H and L pressure impulse lines shall run parallel to each other with minimal elevation changes between them.

7.4.4.8 For steam service, install T's at the highest elevations of the H and L impulse lines to allow water filled into the lines during commissioning.

7.4.4.9 Differential head type instruments for flow measurement shall be the same type as used for differential pressure measurement. The appropriate NEMA classification for the area in which it resides shall be met.

7.4.5 Venturis

7.4.5.1 The venturi shall be able to meet the pressure and temperature ratings of the piping run in which it resides.

7.4.5.2 The material of the venturi shall be compatible with the fluid to be measured and the piping classification of the run in which it is to reside.

7.4.5.3 Venturis shall be oriented such that their taps are at the side of the pipe.

7.4.5.4 Differential sensing lines shall have horizontal sections slope down towards the dP cell transmitter.

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7.4.5.5 The H and L pressure impulse lines shall run parallel to each other with minimal elevation changes between them.

7.4.5.6 For steam service, install T's at the highest elevations of the H and L impulse lines to allow water filled into the lines during commissioning.

7.4.5.7 Differential head type instruments for flow measurement shall be the same type as used for differential pressure measurement. The appropriate NEMA classification for the area in which it resides shall be met.

7.4.6 Vortex Shedding Flowmeters

7.4.6.1 Vortex shedding flowmeters shall be able to meet the pressure and temperature ratings of the piping run in which they reside.

7.4.6.2 The material of the vortex shedding flowmeter shall be compatible with the fluid to be measured and the piping classification of the run in which it is to reside.

7.4.6.3 Vortex shedding meters may be either be the wafer or flanged type. They should be installed with the straight runs upstream and down per the manufacturer's recommendations.

7.4.6.4 Do not use vortex shedding flowmeters in applications where vibration is present.

7.4.7 Paddle Wheel Flowmeters

7.4.7.1 The paddle wheel flowmeter's transmitter may be integral or remotely mounted. The appropriate NEMA classification for the area in which it resides shall be met.

7.4.7.2 Paddle wheel flowmeters must be removable for servicing on "live" systems and therefore must be mechanically bypassed.

7.4.8 Pitot Tube Type Flowmeters

7.4.8.1 Pitot tubes shall be applied per manufacturer's recommendations and have the associated dP cell transmitter calibrations sized per the manufacturer's recommended formulas.

7.4.8.2 The dP cell transmitter will have the appropriate NEMA classification for the area in which it resides.

7.4.8.3 The H and L pressure impulse lines shall run parallel to each other with minimal elevation changes between them.

7.5 Level Measurement.

7.5.1 General Requirements

- 7.5.1.1 The appropriate NEMA classification for the area in which it resides shall be met.
- 7.5.1.2 Level instruments shall be able to meet the pressure and temperature ratings in which they reside and be chemically compatible with the process fluid.
- 7.5.1.3 Switches shall be rated at a minimum of 5 amps at 120 Vac or 24 Vdc. The switch shall achieve a rating that meets the service for which they are intended.
- 7.5.1.4 Level switches shall not utilize a mercury switch.
- 7.5.1.5 The transmitters are recommended to be two wire and provide a 4-20 mA dc linear output for a 0-100% level span.

7.5.2 Differential Pressure Level Instruments

- 7.5.2.1 Differential pressure type level instruments may be used for level measurement on liquid with constant density and can be used for most services except for level spans of less than 20 inches or 0.5m and liquid-liquid interface service. When differential pressure type instruments are to be mounted at or below the taps, they shall be furnished with zero elevation and suppression adjustments.
- 7.5.2.2 For open tank or vessel, a 3inch 150 ANSI flanged mounted liquid level transmitter shall be specified for standard services.
- 7.5.2.3 When electronic level transmitter is specified to measure the level of a closed tank with corrosive or viscous fluids, a differential pressure cell transmitter with remote diaphragm seals sized to the tank. The length of capillary tubing shall be sufficiently long enough to connect the transmitter.

7.5.3 RF/Capacitance Level Instruments

- 7.5.3.1 RF/Capacitance type level transmitters shall be used in services with fluids of low viscosity and solid content.
- 7.5.3.2 A stilling well shall be provided for the probe.
- 7.5.3.3 Remote electronics shall be two wire transmitter which provides a 4-20 mA DC linear output for a 0-100 percent level span of switch contacts for point level trips.

7.5.4 Ultrasonic Level Instruments

- 7.5.4.1 Ultrasonic type level transmitters shall be used in services where non-contacting parts are desirable such as for acid caustic, and corrosive services.
- 7.5.4.2 The unit shall be positioned such that it has an uninterrupted path to the liquid target, preferably in a stilling well.
- 7.5.4.3 The unit shall be installed such that the transducer can emit a sonic pulse perpendicular to the liquid surface.

7.5.5 Float Level Instruments

- 7.5.5.1 Float type level instruments shall generally be used for non-modulating, point type operations such as alarming, shutdowns, etc. A case of modulating service would be self-contained level (fill) control of tanks or cooling tower basins.
- 7.5.5.2 Common applications would be sump alarms, high level or low level cut-outs, etc.
- 7.5.5.3 The switch shall be either SPDT or DPDT and achieve a rating that meets the service for which they are intended.

7.5.6 Weight Type Level Instruments

- 7.5.6.1 Weight type level instruments shall be used in areas where differential pressure, ultrasonic, or capacitance present operational difficulties.
- 7.5.6.2 Load cells shall be constructed of stainless steel and mounted in accordance with manufacturer's recommendations.
- 7.5.6.3 The load cells shall be selected in accordance with the operating load each cell is expected to see.
- 7.5.6.4 Connecting cables shall use acceptable insulation or armoring.
- 7.5.6.5 Remote electronics shall provide a 4-20mA dc linear output for 0-100% span. A local indicator shall be provided to display actual tank weight or 0-100% level.

7.6 Pressure/Differential Pressure Instrument

7.6.1 General Requirements.

- 7.6.1.1 All absolute/gauge pressure transmitters or switches shall be installed with a Swagelok tee and a ¼ turn Whitey valve on the process side for calibration. The tee shall be mounted between the Whitey valve and the transmitter.
- 7.6.1.2 All differential pressure transmitters or switches shall be installed with a two Swagelok tees and three Whitey valves. For each impulse line, the tee shall be mounted between the Whitey valve and the transmitter. To allow for pressure balancing, one Whitey valve shall connect to both impulse lines between the tees and the transmitter.
- 7.6.1.3 Impulse lines shall typically be ½ inch 316L stainless steel in all applications.
- 7.6.1.4 All lines for steam service shall be installed with a coil in the sensing line. They shall be installed with a coil in the sensing line to trap condensate and protect the sensing element.
- 7.6.1.5 The appropriate NEMA classification for the area in which it resides shall be met.
- 7.6.1.6 Pressure shall be able to meet the pressure and temperature ratings in which they reside and be chemically compatible with the process fluid.

7.6.2 Pressure/Differential Pressure Transmitters

- 7.6.2.1 Electronic transmitters shall be two-wire transmitters with 4-20mA output. The transmitter shall be provided with 316 stainless steel wetted parts and isolating diaphragm or other acid-proof materials per process requirement.
- 7.6.2.2 It shall have local indicator display.
- 7.6.2.3 The over pressure shall be two times measuring range.
- 7.6.2.4 Fab Cleanroom pressure and exhaust pressure shall be measure by Differential pressure sensor. The sensor shall measure the difference between Cleanroom pressure and outdoor air, via ZRP (zero reference pressure pipe). The ZRP provides a single common "zero" point that all areas can refer.
- 7.6.2.5 Accuracy shall be 0.15% of full reading span.

7.6.3 Pressure Switches

- 7.6.3.1 Pressure switches for direct connected process and utility services shall be diaphragm type with materials suitable for the services. Switch elements shall provide field adjustable setpoint and be rated as a minimum at 5 amps at 120/240 Vac, 60Hz. The enclosure shall meet the required electrical classification.
- 7.6.3.2 The output signal should be dry contact, SPDT.
- 7.6.3.3 The overload range should be 1.3 times of measuring range.

7.6.4 Pressure Gauges

- 7.6.4.1 Utilize ½ inch threaded with ¼ turn ball valve at process connection point. All gauge connections will be ¼ inch. Use reducer when necessary.
- 7.6.4.2 Diaphragm seals shall be used for prevention of corrosion or clogging of pressure gauges in normal process areas and the size of diaphragm seals shall normally be 1" NPT threaded and 2" flanged.
- 7.6.4.3 Cases for gauges in utility process areas shall be stainless steel with a retainer ring.
- 7.6.4.4 Dials shall be corrosion resistant and white scale with black figures. Dials or pointers shall be field adjustable for zero alignment.
- 7.6.4.5 All direct-connected gauges in process service shall be specified for an accuracy of at least 1% of full scale.

7.7 Temperature Instrument

7.7.1 General Requirement

- 7.7.1.1 Standard 3-wire RTD with transmitter shall be used for all process system temperature control and monitoring. These shall be provided as a single assembly for piping systems.
- 7.7.1.2 Industrial grade combination temperature and humidity sensors shall be used for all space monitoring.
- 7.7.1.3 The output for each transmitter shall be 4-20mA dc for 0 to 100% in engineering units of the calibrated range. Ranges shall be selected to cover the normal operating temperatures experienced, to place the control setpoint in the middle third of the range, and

to keep the range narrow for percent of span accuracy.

- 7.7.1.4 Direct mounted dial thermometers shall normally be bi-metallic type, hermetically sealed, non-reset, heavy duty, every angle with minimum 3.5 inch dial and stainless steel case, and ± 0.1 °C accuracy or better.
- 7.7.1.5 Temperature switches shall be filled-system bulb type or vapor expansion type. They shall have field adjustable setpoint, one DPDT or 2 SPDT dry contacts rated at minimum 5 amps @120/240 Vac, 60Hz. Switch enclosures shall meet the required electrical classification.
- 7.7.1.6 The appropriate instruments shall be able to meet the pressure and temperature ratings in which they reside and be chemically compatible with the process fluid.

7.7.2 Thermowells

- 7.7.2.1 Temperature wells are required for all resistance elements, bi-metallic, thermocouple, thermometers and temperature test services. They shall be constructed for interchangeability for all standard applications. Material shall be typically ANSI Type 316 stainless steel (unless otherwise noted) machined from bar stock in a tapered or stepped configuration, capable of withstanding operating temperature and pressure 1000 °F or 538 °C and 2000 psig.
- 7.7.2.2 Thermowell process connections shall be $\frac{3}{4}$ inch NPT.
- 7.7.2.3 Instrument connection size for all thermowells shall be $\frac{1}{2}$ inch NPT.
- 7.7.2.4 Thermowells in combining streams shall be a minimum of 10 pipe diameters downstream of the junction for proper mixing.

7.7.3 Resistance Temperature Detectors (RTD)

- 7.7.3.1 Platinum RTD shall have an electrical resistance of 100 ohm at 0 °C and a temperature coefficient of resistance (TCR) of 0.00385 ohm/ohm/ °C between 0 and 100 °C, based on DIN/IEC 60751.
- 7.7.3.2 RTD connection heads shall be of the corrosion resistant metallic weatherproof type with terminal blocks enclosed.
- 7.7.3.3 RTD transmitters shall be capable of sensing 3-wire RTD's. They shall be two wires devices for the transmission side, transmitting 4-20 mA dc for the

calibrated range. Transmitter shall be integral or remote from sensors based on applications. If the measurement point is inaccessible, the transmitter shall be remote from the probe to allow its placement in an accessible location.

8.0 Nameplates

- 8.1 Each instrument shall have unique tag number.
- 8.2 Each instrument shall have permanent nameplate with tag number attached to the instrument.

9.0 Installation

- 9.1 Install instruments and control valves in accordance with the approved location drawings and manufacturer's installation instructions. Install instruments level, plumb and accessible from grade, platforms, ladders, or catwalks. Install locally mounted indicating transmitters and gauges faced toward the normal operating aisle, capable or being read from normal line of sight.
- 9.2 Install brackets shown for attachment to walls, columns, masonry, or structural steel so as not to obstruct any access or egress.
- 9.3 Group instruments where practical and mount in locations so as not to block motors or equipment required to be pulled for maintenance or checkout.
- 9.4 Mount instruments level and plumb, rigidly supported in a manner that eliminates transmission of vibration to adjoining structures, components, walls, or cabinets, free from interference of piping and electrical conduit. Services brought to the instruments must not prevent the installation or removal for maintenance purposes. Process tubing routed to the instrument must not block access to the instrument.
- 9.5 Do not exceed process temperature limit, ambient temperature limit, and storage temperature limit.
- 9.6 Install instrument devices in accordance with appropriate installation procedures to ensure the manufacturers' published accuracy of the devices.
- 9.7 Complete necessary process connections between the first process block valves and the instrument. Route vents and drains from instrument process piping to the proper vent headers or sewers as required for environmental reasons.
- 9.8 Instrument Connections
 - 9.8.1 Install Pressure and Differential Pressure Transmitters:
 - 9.8.1.1 Mount as close as possible to the measurement points and maintain accessibility for maintenance.

- 9.8.1.2 Install a test port at the transmitter.
- 9.8.1.3 Install three valve manifolds with differential pressure devices.
- 9.8.1.4 Install chemical diaphragm seals on non-pure process measurements.
- 9.8.1.5 Provide filling fluid suitable for prevention of volatile damage in case of rupture of the diaphragm into the process line.
- 9.8.1.6 Provide fill fluid proven safe for manufacturing processes.
- 9.8.1.7 For Liquid and steam services:
 - 9.8.1.7.1 Mount transmitters below the process taps.
 - 9.8.1.7.2 Slope tubing runs toward the transmitter.
 - 9.8.1.7.3 Use filling tees with fluid heavier than process liquid or condensate where required to eliminate high temperature on the diaphragm.
 - 9.8.1.7.4 Install a trap leg at the transmitter for blowdown of impulse piping.
 - 9.8.1.7.5 Set the filling tees directly across from the taps to equal elevations.
 - 9.8.1.7.6 Place the transmitter in service being careful not to blow the filling liquid into the line.
 - 9.8.1.7.7 Install tubing with equal lengths run to differential transmitters.
- 9.8.1.8 Gas and Liquid with solid in suspension service:
 - 9.8.1.8.1 Mount transmitters above the process taps.
 - 9.8.1.8.2 Slope tubing runs towards the process tap connection.
 - 9.8.1.8.3 Use clean purge fluid in liquids with solids in suspension to prevent clogging the impulse lines.
- 9.8.2 Install chemical diaphragm seals and filled system capillary tubing for maximum support and protection. Coil and secure excess capillary at the instrument. Do not crimp, cut, or bend sharply.

9.9 Instrument Air System:

- 9.9.1 Tie into the instrument air system at the header shutoff valves and extend piping to areas requiring control air. Provide one air drop to handle a maximum 16 branch feeds. Coordinate piping with other contractors.
- 9.9.2 Provide instrument air circuit boxes with isolation valves for instrument air branch feeders where more than six branch feeders are required from an instrument air drop.
- 9.9.3 Control piping shall be:
 - 9.9.3.1 With block valve at instrument.
 - 9.9.3.2 Install and support tubing of required size in a neat and professional manner. Avoid use of short lengths of tubing in long tubing runs. Bend tubing wherever possible to reduce number of joints and possibility of leaks. All bends in metal tubing should be made with a tool designed for that purpose to prevent kinks or flattening. Use standard tubing unions to connect tubing and always use tubing bender to bend tubing.
 - 9.9.3.3 Install multiple runs of tubing in tubing tray or supported in protecting channel. Support single runs of tubing independently. Install tubing so that overlapping or interweaving is kept to minimum and arranged so that connections are readily accessible for tightening without distorting and bending lines.
 - 9.9.3.4 Do not route tubing, regardless of location, against cold ducts, hot or cold piping, equipment, or under insulation.
 - 9.9.3.5 Test piping system under working pressure plus 50% for a period of 24 hours with loss of pressure not exceeding 10%.
 - 9.9.3.6 Install bulkhead-type union connections for tubing penetrating panels, plenums, ducts, or airshaft walls.
 - 9.9.3.7 Main air piping must be ½ inch minimum, and branches ¼ inch minimum.
 - 9.9.3.8 Provide plastic tubing connection to instrument from isolation block valve to prevent vibration except in corrosive areas where stainless tubing must be used.

9.10 Cleanroom Temperature & Humidity Sensor installation

- 9.10.1 Ceiling mounted Temperature and Humidity sensors shall be used in each Temperature Control Zone.
- 9.10.2 Temperature sensor probe is recommended to install 100mm under FFU, and 200mm in FFU. This sensor location allows the sensor to sense adequate mixing of the FFU supply air (from Dry Cooling Coil and interstitial plenum) mixed with the Cleanroom heat sources. The other benefits are
 - a. The sensor shall provide repeatability of results in dry cooling coil control.
 - b. The sensor shall be located away from AMHS track, and other obstructions and heat sources such as exhaust ductwork, utility piping, etc.
- 9.10.3 The transmitter module shall be mounted up at interstitial plenum along catwalk or grating, to allow easy access for calibration and troubleshooting. The probe shall have excess cable length of minimum 10m to allow for future relocation of the probe.
- 9.10.4 The mounting method can refer to figures below:

Figure 1a –Cleanroom T&H probe mounting location

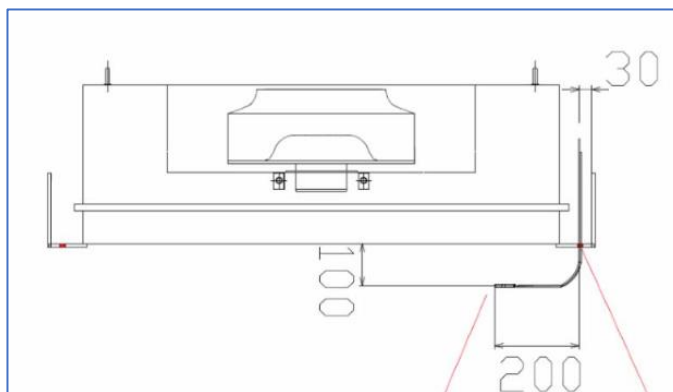


Figure 1b –Recommended Cleanroom T&H probe mounting location (Photo taken)



9.11 Zero Reference Pressure.

9.11.1 Fab Cleanroom and Exhaust differential pressure shall be relative to outside ambient pressure via zero reference pressure (ZRP) pipe.

9.11.2 The ZRP pipe shall have the following, but not limited to, requirements:

- i. Th ZRP piping shall be leak-tight Stainless steel SS304
- ii. There shall be isolation valve between levels.
- iii. There shall be isolation valve for each instrument connection.
- iv. All end connection shall be provided with end cap.
- v. Leak test shall be executed after installation.

9.11.3 The should be ZRP box and ZRP buffer tank before connection to atmosphere. The recommended design are shown in figures below:

Figure 2a – ZRP pipe terminal elements.

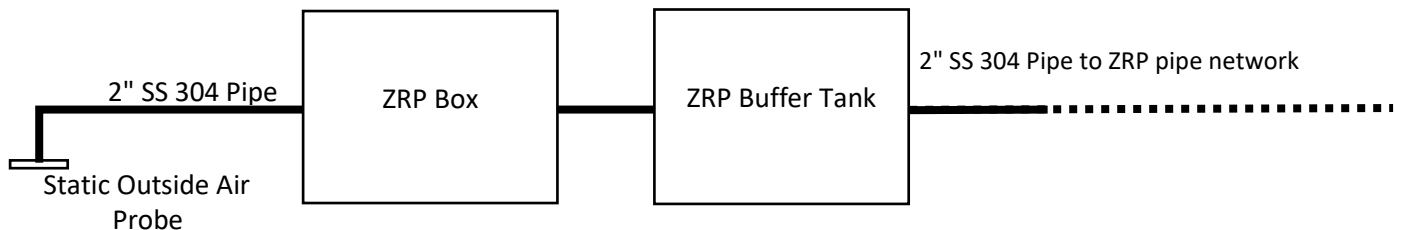


Figure 2b – ZRP Buffer Tank recommended detail

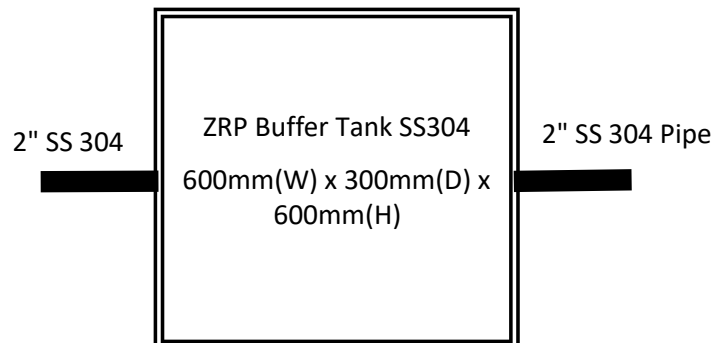
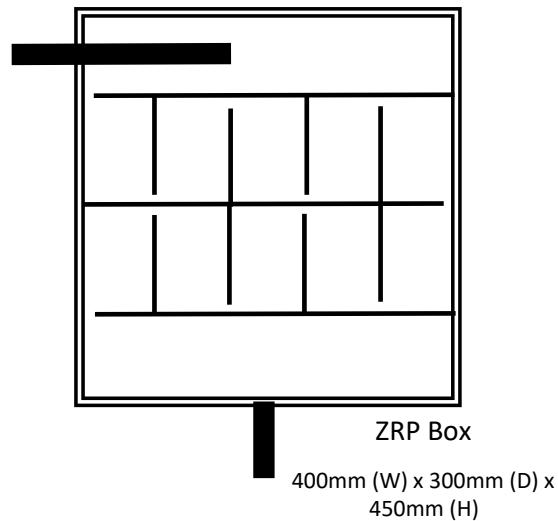


Figure 2c –ZRP Box recommended detail

- 9.11.1 The static outside air probe shall be installed in a stable location with outdoor ambient, and protected from potential effects of weather conditions including wind, rain, etc.

9.12 Field Wiring Terminations

- 9.12.1 Provide field terminations and conduit knockouts for control/instrumentation wiring. Field termination wiring must have designated instrument tag numbering
- 9.12.2 Provide cut, punched, or drilled and smoothly finished with round edges panel cutouts/knockouts.
- 9.12.3 Provide separate conduit entry for each power feeder circuit.
- 9.12.4 Ground signals requiring grounding within the panel. Fold back and place under the heat-shrink tubing the field end of the cable without being grounded.
- 9.12.5 Cover conductor shield/drain wires with clear tubing at the panel and ground the drain.
- 9.12.6 Restrain control panel wiring with plastic ties or ducts. Secure hinge wiring at each end so that any bending or twisting will be around the longitudinal axis of the wire and protect the bend area with a sleeve.
- 9.12.7 Arrange wiring neatly, and provide abrasion protection for wire bundles that pass through holes or across edges of sheet metal.
- 9.12.8 Use manufacturer's recommended tool with the proper-sized anvil for crimp terminations. No more than two wires may be terminated in a single crimp lug and no more than two lugs may be installed on a single screw terminal.

9.12.9 Splicing or tapping of wiring is not acceptable except at device terminals or terminal blocks.

9.12.10 Calibrate or provide factory calibration for instrumentation provided on control panels.

10.0 Instrumentation commissioning

10.1 General Requirement

- i. Verify instrument and its installation comply with Contract Documents and manufacturer's recommendations.
- ii. Verify instrument signal and power wiring comply with Contract Documents and manufacturer's recommendations.
- iii. Verify instrument air and control air piping comply with Contract Documents and manufacturer's recommendations.
- iv. Provide portable compressed air source for bench testing and for air calibrations required prior to final installation of instrument air.
- v. Recommended calibration forms are available in Appendix C for Conductivity, PH sensor.
- vi. Recommended instrument loop check form is available in Global Facilities - Design & Construction Standard - I&C - Facility Management and Control System (FMCS) Appendix C.

10.2 Control Valves/Dampers:

- i. Calibrate and adjust positioners and IP transducers.
- ii. Verify proper control action (i.e., air to open/air to close) and proper failure position on loss of air, loss of power, and loss of signal.
- iii. Adjust limit switch settings.
- iv. Adjust opening and closing speeds and travel stops.
- v. Perform five-point calibration check.
- vi. Stroke control valves by means of associated controller.

10.3 Pressure and Differential Pressure Instruments:

- i. Perform five-point calibration test in percent of input versus output.
- ii. Verify zero point under pressure and re-zero if necessary.

10.4 Temperature Transmitters:

- i. Verify element is in correct position and sufficient length to provide accurate measurement of variable.
- ii. Verify correct five-point input versus output using decade precision resistance box or potentiometer.

10.5 Pressure Switches:

- i. Verify correct switch operation at setpoint (make on increasing pressure, break on increasing pressure, make on decreasing pressure, break on decreasing pressure).
- ii. Verify reset (automatic or manual) and deadband if applicable.

10.6 Level Switches

- i. Determine correct depth at which the switch is supposed to operate.
- ii. Manually position the switch at this level and verify correct operation.
- iii. In the case of pressure actuated level switches, simulate pressure corresponding to level setting and verify operation.

11.0 Document Control

11.1 **Approval:** GLOBAL_FAC_GFTT_APPR

11.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F2_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

11.3 Final Viewer Access

All Micron sites Facilities Team Members

11.4 Retention

Adherence to the requirements specified in

[**Global Facilities - Document Control Procedures**](#)

11.5 Review

Biennially (two years)

12.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	03/08/21

Appendix A: Deviation List

Any deviation from this specification shall be submitted for approval during vendor evaluation. Deviation list format shall be the following:

Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT

Appendix B: Instrument List Format

Item No.	System	Tag No	P&ID No.	Rev. No.	Description	Type	Process Fluid	Normal Operation Range	Calibrated Range / Setpoint	Accuracy	Units	LL	L	H	HH	Power Supply	Analog / Digital Signal (2 / 4 wire)	Output (4-20mA, Pulse, Contact, Comm. bus)	Process Connection	Pipe Size (mm)	Wetted Part Material	Sensor Insertion Length (mm)	Indicator (None, Local, Remote)	IP Rating (Sensor)	IP Rating (Transmitter / Switches)	Specification Reference	Make / Brand	Transmitter / Switch Model No.	Element / Sensor Model No.	Other Accessories Model No.	Country of Origin	Package Vendor	Procurement Status	Year of Installation	Remarks
1																																			
2																																			
3																																			
4																																			
5																																			

Appendix C: Analyzer Commissioning Checklist

Conductivity Transmitter Calibration

Tag No					Inst – Range	
SCADA Tag					Cal – Range	
Description					Output Signal	
Model					Supply	
Sensor Model					Make	
Equipment Name						
Location						
Before Calibration						
Standard Buffer solution	Measured Output Current (4-20mA)	Local Display Reading	Acceptable Accuracy	Calibration Required (Yes/No)	Remark	
After Calibration						
Standard Buffer solution	Measured Output Current (4-20mA)	Local Display Reading	Acceptable Accuracy	Calibration Required (Yes/No)	Remark	
		Local	SCADA	Cell Constant		
As Found Process Conductivity				As Found Cell Constant		
As Left Process Conductivity				As Left Cell Constant		

	Tested by	Witnessed by	Approved by
Company, Signature, Name, Date			

pH Transmitter Calibration

Tag No				Inst – Range	
SCADA Tag				Cal – Range	
Description				Output Signal	
Model				Supply	
Sensor Model				Make	
Equipment Name					
Location					
Before Calibration					
Standard Buffer solution	Measured Output Current (4-20mA)	Local Display Reading	Acceptable Accuracy	Calibration Required (Yes/No)	Remark
After Calibration					
Standard Buffer solution	Measured Output Current (4-20mA)	Local Display Reading	Acceptable Accuracy	Calibration Required (Yes/No)	Remark
		Local	SCADA	Cell Constant	
As Found Process pH Reading				As Found Slope Value	
As Left Process pH Reading				As Left Slope Value	

	Tested by	Witnessed by	Approved by
Company, Signature, Name, Date			